PLEUR-EVAC®

THE SAHARA® SYSTEM FEATURES:

(A) **EASY-TO-READ VISUAL INDICATOR AND SUCTION CONTROL**

(B) **PATIENT AIR-LEAK METER:** *quantifies the size and progress of air leak*

(C) **FLOOR STAND:** *helps prevent unit tipover*

SETUP INSTRUCTIONS:

1. Connect patient drainage tube **(number 1, right)** to thoracic catheter to establish one-way seal for patient protection.
2. For air-leak diagnostics, add pre-packaged 20 mL sterile fluid to air leak meter through the needleless injection site **(number 2, right)**. Fill to the 20 mL line.
3. If suction is required, connect the suction source to suction port **(number 3, right)**. Increase source suction until orange float appears in window.

SAHARA® FLUID MANAGEMENT SYSTEM

Dry Suction, Dry Seal

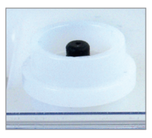
SAHARA® SYSTEM FEATURES AND BENEFITS

DRY SUCTION CONTROL CHAMBER

The dry suction control delivers from -10 to -40 cm of water suction with calibrated control and accuracy. The suction control feature functions as a true regulator: as fluctuations occur in source suction or airflow from a patient leak, the desired imposed suction level is automatically maintained without the need for clinical intervention. The orange float appears in the window when prescribed suction is attained, eliminating guesswork. Setup is quick and easy, with no need to fill suction chamber with water.

TO INCREASE SUCTION
The Pleur-evac Sahara® Unit is preset at -20 cm of water. Rotate the red dial to the prescribed suction level. The dial clicks in place. Increase source suction until the orange float appears in the indicator window. Source suction must be capable of delivering a minimum of 6 liters per minute (LPM) airflow.

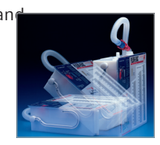
TO DECREASE SUCTION – MANUAL HIGH NEGATIVITY RELIEF VALVE*



Rotate the red dial to the prescribed suction level. The dial clicks in place. When suction setting is changed from a higher to a lower setting, the patient negativity may remain at the higher level unless the negativity is relieved. Use the filtered high negativity relief valve located at the top of the unit to reduce patient negativity to desired level. Depress the button to manually relieve excess negativity.

GRAVITY DRAINAGE
If gravity drainage is prescribed, suction port should remain uncapped and free of obstructions.

ONE-WAY VALVE



The Sahara® System one-way valve allows air to exit from the pleural space and prevents air from entering. The one-way valve sets up quickly and easily with no need to fill a water seal. The one-way valve protects the patient seal, even when the unit is tipped over. There is no need to replace the unit if tipped, as fluid stays in its own chamber.

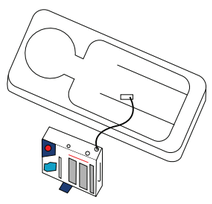
MONITORING PATIENT AIR LEAKS

The patient air leak meter quantifies the size of an air leak from 1 (low) to 7 (high). The higher the numbered column through which bubbling appears, the greater the magnitude of the air leak. Observe bubbling at the bottom of the columns of the air leak meter. By documenting the number, the clinician can monitor air leak increase or decrease. The fluid in the air leak meter is used for air leak detection and is not a water seal.

AUTOMATIC POSITIVE PRESSURE RELIEF VALVE/ HIGH NEGATIVITY RELIEF
The automatic positive pressure relief valve automatically opens with increases in positive pressure if the suction tube or port is blocked, minimizing the risk of tension pneumothorax. Negative pressure is limited to approximately -50 cm of water.

* **CAUTION:** If suction is not operative, or if operating on gravity drainage, depressing the high negativity relief valve can reduce negative pressure within the collection chamber to zero (atmosphere), with the resulting possibility of a pneumothorax.

COLLECTION CHAMBER



The long patient tubing connects to the thoracic catheter. Drainage flows into the graduated collection chamber. The write-on surface allows for easy measurement and recording.

POSITIONING THE PLEUR-EVAC® SAHARA® SYSTEM
Position Sahara system below the patient's chest at all times. Avoid positions that create loops in patient tubing.

TO CLAMP OR NOT TO CLAMP
Chest tubes should not be clamped unless changing the unit or ordered by a physician.

If clamping is necessary, clamp near the color-coded autotransfusion system (ATS) connectors.

RECORDING PATIENT DRAINAGE
Mark the level of drainage with date and time of measurement on white write-on surface. Sahara units feature an easy-to-read calibrated measurement scale: 2cc increments up to 200cc, 10cc increments up to 2000cc.

SAMPLING PATIENT DRAINAGE OR REPLACING UNITS USING THE COLOR-CODED ATS CONNECTORS
The needleless sampling port in the connector allows clinicians to obtain fresh samples of patient drainage. Use a standard luer lock syringe for withdrawing samples. Color-coded ATS connectors and clamp-on drainage tube allow for quick and easy Sahara unit replacement.

NURSING CONSIDERATIONS AND TROUBLESHOOTING*

DRY SUCTION CONTROL

IS THE DIAL SET AT THE PRESCRIBED SUCTION?
Turn the dial to click into the correct suction setting (-20 cm water suction is most common for adults).

IS THE ORANGE FLOAT IN THE INDICATOR WINDOW?
The orange float indicates that the desired suction has been achieved. This visual indicator replaces the bubbling noise of a water suction control system.

The suction source must be capable of delivering a minimum of 6 liters per minute (LPM) of airflow. The suction source must deliver greater than 6 LPM when multiple chest drains are connected to a single source.

If the orange float falls out of the indicator window, the wall suction source must be increased.

If the orange float still does not appear, check to make sure that all connections are secure and the suction tubing is not kinked, clamped, or occluded. You may need to bypass the wall canister. When multiple drains are Y-ed together, connect the Y close to the suction source rather than closer to the chest drainage units.

DOES THE WATER RISE IN THE SMALL ARM OF THE AIR LEAK METER WHEN THE DRY SUCTION SETTING IS LOWERED?
This is normal. It simply reflects the previous higher setting. If the patient does not have an air leak, vent the excess negativity by depressing the high negativity relief valve.

NEGATIVE PRESSURE INDICATOR

IS THE NEGATIVE PRESSURE INDICATOR VISIBLE?
During gravity drainage, the indicator may intermittently indicate a negative pressure in the collection chamber with patient respiration. During suction drainage, the pressure indicator should indicate a negative pressure continuously.

CAUTION: If the negative pressure indicator does not show "yes," as described: 1) check patient connections for leaks, 2) check tubing connections on the unit. If all connections are secure and the "yes" does not appear, replace the unit. The negative pressure indicator does not confirm drainage tube patency. Routinely check the drainage tube patency.

AIR LEAK METER

IS THERE BUBBLING?
Identify the source of the air leak:
- Check and tighten connections.
- Check the tubing for leaks using progressive clamping method with booted (or padded) chest tube clamp.
- If leak is in the tubing, replace the unit.
- If the leak is at the insertion site, remove the chest tube dressing and inspect the site. Make sure the drainage ports have not pulled out beyond the chest wall. If you can not see or hear any obvious leaks at the site, the leak is from the pleural cavity. Replace the dressing.
- Check patient history. Would you expect an air leak?

IS THE BUBBLING CONTINUOUS OR INTERMITTENT?
Note the pattern of the bubbling. If it fluctuates with respiration (i.e., occurs on exhalation in a patient breathing spontaneously), the most likely source is the pleural cavity.

Notify doctor of any new, increased, or unexpected air leaks that are not corrected by the above actions.

Document the magnitude of a patient air leak using the air leak meter. The higher the numbered column through which the bubbling occurs, the greater the degree of air leak.

DRAINAGE TUBES

ARE ALL CONNECTIONS SECURELY TAPED OR BANDED?
Reconnect any loose connections and tape securely; assess for a new or increased air leak. Notify doctor if new or increased air leak present.

IS THE CLAMP OPEN?
The tube should be left unclamped unless:
a. You are changing the unit. Clamp only briefly.
b. Specifically ordered by the doctor.

Do not clamp the chest tube during transport or ambulation unless specifically ordered. Clamping the chest tube in patients with an air leak increases the possibility of pneumothorax.

COLLECTION CHAMBER

IS DRAINAGE BLOODY, STRAW-COLORED, OR PUS-COLORED?
Document findings. Notify doctor if drainage character changes significantly, i.e., straw-colored drainage at last check is now bloody.

HAS THE RATE OF DRAINAGE SUDDENLY INCREASED OR DECREASED?
Sudden hemorrhage in a postoperative cardiac patient is likely caused by a ruptured suture line or blown graft. The patient can lose 1,000 to 1,500 mL of blood in a matter of minutes. Immediately alert surgeon and prepare for return trip to O.R.

HAS DRAINAGE STOPPED SUDDENLY?
A sudden (not gradual) cessation of drainage in the patient with mediastinal tubes can be caused by accumulated clotted blood occluding the tube. This can lead to life-threatening cardiac tamponade. To keep the tubes patent, or to dislodge clots, gently milk the tube.

If the patient's condition is deteriorating rapidly, follow the emergency procedures of milking/stripping to dislodge clots.

If the patient appears stable, make sure the unit is low enough so gravity can assist drainage: raise the bed, lower the Sahara® unit, or turn the patient on his/her affected side.

Check tubing for kinks or bends. Make sure tube is not clamped.

** This is a trouble shooting guide only. Please refer to the Instructions For Use for full operating and set-up instructions.*

Teleflex, Pleur-Evac and Sahara are trademarks or registered trademarks of Teleflex Incorporated or its affiliates. © 2015 Teleflex Incorporated. All rights reserved. MC-000098

Teleflex®

TELEFLEX 3015 Carrington Mill Boulevard, Morrisville, NC 27560
Toll Free: 866.246.6990 Phone: +1.919.544.8000
TELEFLEX.COM